

Action Myoclonus–Renal Failure Syndrome (AMRF): A Comprehensive Disease Characteristics Report

Disease: Action Myoclonus–Renal Failure Syndrome **MONDO ID:** MONDO:0009699 · **OMIM:** #254900 (EPM4) · **Orphanet:** ORPHA:163703 **Category:** Mendelian, autosomal recessive lysosomal disorder **Causal gene:** *SCARB2* (encoding LIMP-2) · HGNC:10534 · Ensembl ENSG00000138760 · UniProt Q14108 · chr4q21.1

Summary

Action Myoclonus–Renal Failure Syndrome (AMRF) is an ultra-rare, autosomal recessive, multisystem lysosomal disorder that unites two seemingly unrelated clinical problems: a **progressive myoclonus epilepsy (PME)** with preserved cognition and a **proteinuric collapsing focal segmental glomerulosclerosis (FSGS)** that progresses to end-stage renal failure. The syndrome is caused by **biallelic loss-of-function variants in *SCARB2***, the gene encoding **lysosomal integral membrane protein type 2 (LIMP-2)**. LIMP-2 is the mannose-6-phosphate-independent sorting receptor that traffics **β-glucocerebrosidase (GCase)** from the endoplasmic reticulum (ER) to the lysosome. When LIMP-2 is absent or misfolded, GCase fails to reach the lysosome, is retained in the ER, and lysosomal GCase activity falls—producing a partial, tissue-specific enzyme deficiency and downstream glycosphingolipid dysregulation.

The mechanistic chain runs from mutation → LIMP-2 loss/ER-retention → GCase mistrafficking → partial lysosomal GCase deficiency → accumulation of the toxic glycosphingolipid **glucosylsphingosine (GlcSph)** and glucosylated cholesterol (rather than bulk glucosylceramide storage) → neuronal and podocyte injury → the branched clinical phenotype of cortical action myoclonus, tremor, ataxia and seizures on the neurological arm, and glomerular collapse with proteinuria and renal failure on the renal arm. Because residual GCase remains high in leukocytes but is severely deficient in fibroblasts and tissue, standard leukocyte enzyme assays are normal—a diagnostic pitfall that distinguishes AMRF from classical Gaucher disease.

Clinically, AMRF typically presents in **adolescence or young adulthood** (tremor onset ~20 years, disabling action myoclonus ~22 years), with proteinuria often detected in childhood-to-early-adulthood and progressing to renal failure within 0–8 years. Cognition is characteristically spared. There is no approved disease-modifying therapy; management is symptomatic (antimyoclonic/antiseizure drugs, dialysis and renal transplantation). Emerging disease-modifying strategies include **substrate-reduction therapy with miglustat** and **vitamin E repletion**, the latter validated in *Scarb2*-knockout mice. Diagnosis rests on the combination of cortical-myoclonus electrophysiology, urinalysis/renal biopsy findings, and molecular confirmation of biallelic *SCARB2* variants.

Key Findings

F001 — AMRF is caused by biallelic loss-of-function mutations in *SCARB2* (LIMP-2)

Two independent 2008 studies established the genetic basis of AMRF. Berkovic and colleagues used homozygosity mapping in three unrelated families to map the disease locus to 4q13-21 and identified *SCARB2/LIMP-2* by microarray expression analysis; mutations were found in all three mapping families plus two additional AMRF families, associated with an absence of *SCARB2* protein. As the authors state, "*The ancestral lysosomal-membrane protein SCARB2/LIMP-2 is responsible for AMRF*" (PMID: 18308289). Independently, Balreira et al. identified a homozygous nonsense mutation at codon 178 (**W178X**), noting that "*A homozygous nonsense mutation in codon 178 of SCARB2 was found in the patient, whereas her healthy parents were heterozygous for the mutation*" (PMID: 18424452). Crucially, the same paper established the protein's function—*SCARB2* encodes LIMP-2, "*the sorting receptor for beta-glucocerebrosidase*." This gene-to-function link is the foundation of all downstream mechanistic understanding of AMRF.

Ontology/annotation: Gene *SCARB2* (HGNC:10534); protein LIMP-2 / SR-B2 (UniProt Q14108); MONDO:0009699; OMIM #254900.

F002 — The renal phenotype is proteinuric collapsing focal segmental glomerulosclerosis

The renal lesion of AMRF is a severe, collapsing variant of FSGS. Berkovic et al. describe AMRF as *"the remarkable combination of focal glomerulosclerosis, frequently with glomerular collapse, and progressive myoclonus epilepsy"* (PMID: 18308289). Balreira et al. reported nephrotic syndrome with *"a strong accumulation of C1q in capillary loops and mesangium of kidney"* (PMID: 18424452), a pattern designated **nephropathy C1q**. Chaves et al. confirmed PME with nephropathy C1q in two siblings (PMID: 21782476). The presence of collapse and podocyte injury points to the podocyte as the principal renal target cell.

Ontology/annotation: HP:0000097 (Focal segmental glomerulosclerosis); HP:0000100 (Nephrotic syndrome); HP:0000093 (Proteinuria); UBERON:0000074 (renal glomerulus); CL:0000653 (podocyte).

F003 — *Limp2/Scarb2* knockout mice recapitulate multisystem AMRF-relevant pathology

The mouse model predates the human gene discovery and is central to mechanistic understanding. Gamp et al. showed that LIMP-2-deficient mice develop uni/bilateral hydronephrosis from ureteropelvic junction obstruction, serious hearing impairment (spiral ganglia and hair cell loss, stria vascularis degeneration), and a peripheral demyelinating neuropathy: *"LIMP-2-deficient mice are also characterized by a peripheral demyelinating neuropathy"* (PMID: 12620969). Berkovic's reanalysis of the same knockout revealed intracellular inclusions in cerebral and cerebellar cortex and subtle glomerular changes. More recently, Li et al. showed that *Scarb2*-deficient mice have age-dependent dietary lipid malabsorption and vitamin E deficiency via gut dysbiosis and FXR hyperactivation, and importantly that *"inhibiting FXR or supplementing vitamin E ameliorates the neuromotor impairment and neuropathy in Scarb2 knockout mice"* (PMID: 38635907)—a translationally significant therapeutic lead.

F004 — LIMP-2 loss disrupts glycosphingolipid homeostasis (GlcSph, GlcChol) rather than causing bulk GlcCer storage

A key refinement of the "lysosomal storage" paradigm comes from Gaspar et al., who showed that in *Limp2*^{-/-} mouse tissues the only consistently deficient lysosomal enzyme was GCase, and that *"GCase deficiency in tissues does not correlate with increases in GlcCer; but rather with increases in glucosylsphingosine (GlcSph) and glucosylated cholesterol (GlcChol)"* (PMID: 25111111).

[40639771](#)). This reframes AMRF as a disorder of **toxic glycosphingolipid accumulation** (GlcSph is a bioactive, cytotoxic lyso-lipid) rather than classical macromolecular storage. The same study explains a longstanding diagnostic anomaly: *"residual GCase is remarkably high in leukocytes,"* accounting for the normal leukocyte enzyme assays seen in patients despite tissue-level deficiency.

Ontology/annotation: CHEBI:88431 (glucosylsphingosine); GO:0006687 (glycosphingolipid metabolic process).

F005 — Neurological phenotype: adolescent/young-adult onset tremor, progressive action myoclonus, ataxia, seizures — without dementia

Badhwar et al., in the landmark series of 15 patients from 9 families, quantified the neurological course: *"Tremor (onset 17-26 years, mean 19.8 years, median 19 years) and progressively disabling action myoclonus (onset 14-29 years, mean 21.7 years, median 21 years), with infrequent generalized seizures (onset 20-28 years, mean 22.7 years, median 22 years) and cerebellar features are characteristic"* ([PMID: 15364701](#)). The same series established the renal timeline: *"Proteinuria, detected between ages 9 and 30 years in all cases, progressed to renal failure in 12 out of 15 patients within 0-8 years after proteinuria detection."* Rubboli et al. emphasized the preserved cognition and cortical origin of the myoclonus: *"The main clinical features were adolescent-young adulthood onset, progressive action myoclonus, ataxia, absence of cognitive deterioration and, in most cases, epilepsy"* ([PMID: 22050460](#)), with rhythmic myoclonic jerks at 12–20 Hz resembling postural tremor and a demonstrated cortical origin via EEG–EMG coherence.

Ontology/annotation: HP:0001336 (Myoclonus); HP:0002345 (Action tremor); HP:0001251 (Ataxia); HP:0001250 (Seizure); HP:0007000 (Photosensitive myoclonic seizures); absence of HP:0001268 (Mental deterioration).

F006 — SCARB2 mutations cause a spectrum from AMRF to PME without renal failure; C-terminal variant location influences age of onset

The renal phenotype is not fully penetrant. Rubboli et al. described five Italian PME patients with *SCARB2* mutations but **without renal impairment**: *"We describe the clinical and neurophysiologic features of PME associated with SCARB2 mutations without renal impairment"* ([PMID: 22050460](#)), indicating variable expressivity/incomplete penetrance of the renal arm. Atasu et al., in an in-depth literature review, found a genotype–phenotype correlation: *"only the C terminal localization of the pathogenic variant significantly affected the clinical*

presentation, particularly the age at onset" (PMID: 35346091), while variant type had no major impact on the overall course. Intrafamilial heterogeneity is documented even among siblings sharing an identical truncating variant (p.N45MfsX88).

F007 — Treatment is largely symptomatic; substrate reduction therapy (miglustat) shows promise; vitamin E is a candidate

No disease-modifying therapy is approved. Management is symptomatic: antiepileptic/antiseizure medication (valproate, levetiracetam, piracetam, clonazepam, perampanel) and renal replacement therapy. Badhwar et al. underscore the historical importance of renal support: *"The syndrome was not recognized prior to the advent of dialysis and renal transplantation because of its rapidly fatal course if renal failure is untreated"* (PMID: 15364701). Two lines of evidence support **substrate-reduction therapy (SRT)**: Chaves et al. reported that *"When substrate-reduction therapy, to correct the possible glucocerebroside storage in the cells with glucocerebrosidase deficiency, was administered to one of the siblings, a significant improvement was observed"* (PMID: 21782476); and Quraishi et al. reported that miglustat halted myoclonus progression, resolved dysphagia, and allowed reacquisition of skills (PMID: 34337151). Vitamin E repletion or FXR inhibition ameliorated neuromotor deficits in *Scarb2*-knockout mice (PMID: 38635907).

Ontology/annotation: NCIT candidate terms — Miglustat (glucosylceramide synthase inhibitor / SRT), Valproate, Levetiracetam, Clonazepam, Perampanel, Renal Dialysis, Kidney Transplantation, Vitamin E supplementation.

F008 — LIMP-2 has pleiotropic non-lysosomal roles informing multisystem involvement

Beyond its GCase-sorting role, LIMP-2 has additional functions that may contribute to the multisystem phenotype. Schroen et al. showed LIMP-2 is a component of the cardiac intercalated disc that associates with cadherin; *"these LIMP-2 null mice failed to mount a hypertrophic response to increased blood pressure but developed cardiomyopathy"* (PMID: 17485520). Gonzalez et al. reviewed LIMP-2 as *"a receptor for specific enteroviruses, two unanticipated findings that reaffirm the myriad roles of lysosomal proteins"* (PMID: 24389070) (EV71, coxsackievirus A16), noting ~14 disease-causing SCARB2 mutations known as of 2014.

F009 — *SCARB2* is tolerant of heterozygous loss-of-function (gnomAD), consistent with autosomal recessive inheritance

gnomAD constraint metrics for *SCARB2* (ENSG00000138760, chr4:76,158,737–76,234,536, GRCh38) show the gene is **not haploinsufficient**: pLI = 0.0009, observed/expected LoF (oe_lof) = 0.49 with a LOEUF (90% CI upper bound) of 0.67 (observed LoF = 29 vs expected 58.6); missense Z = 1.60. This tolerance of heterozygous LoF is exactly what is expected for a recessive disease gene where a single functional allele suffices. It is consistent with the clinical observation that carriers are healthy—Balreira's *"healthy parents were heterozygous for the mutation"* (PMID: 18424452). The disease is catalogued as EPM4 (progressive myoclonic epilepsy-4 with or without renal failure), OMIM #254900.

F010 — Diagnosis combines cortical-myoclonus electrophysiology, renal/urinalysis findings, and *SCARB2* molecular testing

AMRF diagnosis is a three-legged stool. **Electrophysiology** shows cortical action myoclonus: giant somatosensory evoked potentials, marked photosensitivity, and 12–20 Hz rhythmic myoclonic jerks with EEG–EMG coherence demonstrating cortical origin (PMID: 22050460). Hotait et al. highlighted distinctive EEG features: *"this report emphasizes the presence of two EEG patterns, fixation-off phenomenon, and bursts of parasagittal spikes exclusively seen during REM sleep that appear to be characteristic of this condition"* (PMID: 33343627). **Renal workup** detects proteinuria on urinalysis and collapsing FSGS on biopsy. **Molecular testing** is the gold standard—Yari et al. exemplify this: *"Genetic analysis identified a homozygous splicing c.423+1 G>A variant in the SCARB2 gene of the proband and his affected sister"* (PMID: 33772352). Critically, enzyme testing must use fibroblasts, not leukocytes: Balreira found *"a normal beta-glucocerebrosidase activity in leukocytes, but a severe enzymatic deficiency in cultured skin fibroblasts"* (PMID: 18424452).

F011 — Prognosis: progressive, disabling, with early mortality

Historically fatal from untreated renal failure, AMRF's prognosis is now driven by relentless neurological decline. Badhwar et al. note the disease *"was not recognized prior to the advent of dialysis and renal transplantation because of its rapidly fatal course if renal failure is untreated"* (PMID: 15364701). Quraishi et al. characterize AMRF as *"a rare, progressive myoclonic epilepsy with early mortality"* (PMID: 34337151). The trajectory is proteinuria (childhood–early adulthood) → renal failure within 0–8 years, and action myoclonus/tremor progressing to

severe disability, dysphagia, and death (often from neurological complications, aspiration, or status epilepticus once renal failure is managed). Cognition is typically preserved throughout (PMID: 22050460).

F012 — Mechanistic basis: AMRF mutations cause ER retention of LIMP-2 and disrupt pH-dependent coiled-coil binding to GCase

The molecular pathology is precisely characterized. Blanz et al. showed that *"All mutations investigated in this study lead to a retention of LIMP-2 in the endoplasmic reticulum (ER) but affect the binding to beta-GC differentially"* (PMID: 19933215); binding occurs through a highly conserved amphipathic coiled-coil domain (segment 145–288), and its disruption abolishes GCase binding. Zachos et al. defined the release mechanism: *"the luminal acidification mediated by the vacuolar (H(+))-ATPase triggers the dissociation of LIMP-2 and GC in late endosomal/lysosomal compartments"* (PMID: 22537104), with a critical histidine residue conferring pH sensitivity. In patient cells, Balreira confirmed GCase mislocalization: *"decreased amounts of beta-glucocerebrosidase, which was mainly located in the endoplasmic reticulum, as assessed by its sensitivity to Endo H"* (PMID: 18424452).

F013 — Anatomical involvement: cerebral/cerebellar cortex, kidney glomeruli/podocytes, peripheral nerve; lysosome the key compartment

Badhwar et al. documented the tissue-level pathology: *"Brain autopsy in two patients revealed extraneuronal pigment accumulation. Renal biopsies showed collapsing glomerulopathy, a severe variant of focal glomerulosclerosis"* (PMID: 15364701). Berkovic's knockout reanalysis localized storage pathology: the mice *"showed intracellular inclusions in cerebral and cerebellar cortex, and the kidneys showed subtle glomerular changes"* (PMID: 18308289). The cortical origin of myoclonus localizes to sensorimotor cortex; the peripheral nervous system shows axonal/demyelinating polyneuropathy; and the subcellular locus is the lysosome (LIMP-2 being a lysosomal integral membrane protein).

Ontology/annotation: UBERON:0000956 (cerebral cortex); UBERON:0002129 (cerebellar cortex); UBERON:0002113 (kidney); UBERON:0000074 (renal glomerulus); UBERON:0000044 (peripheral nerve); GO:0005764 (lysosome); GO:0005783 (endoplasmic reticulum); CL:0000653 (podocyte); CL:0000540 (neuron).

F014 — Epidemiology and etiology: ultra-rare, purely monogenic autosomal recessive; consanguinity increases risk

AMRF is ultra-rare (Orphanet ORPHA:163703; prevalence <1/1,000,000). Only ~4 patients were known before 2004; Badhwar's landmark series expanded this: "*We now describe 15 individuals with AMRF from five countries*" (PMID: 15364701). Since 2008, several dozen SCARB2-related cases have been reported worldwide, many from consanguineous families—Ekmekci et al. note "*This study examines a consanguineous family with multiple members presenting myoclonic epilepsy*" (PMID: 37529812). The etiology is entirely genetic (biallelic pathogenic SCARB2 variants); no environmental, infectious, toxic, or lifestyle cause is implicated. Sex ratio is ~1:1, as expected for an autosomal recessive disorder.

F015 — Prevention is limited to genetic counseling, carrier/cascade testing, and prenatal/PGD; secondary/tertiary prevention targets renal and nutritional complications

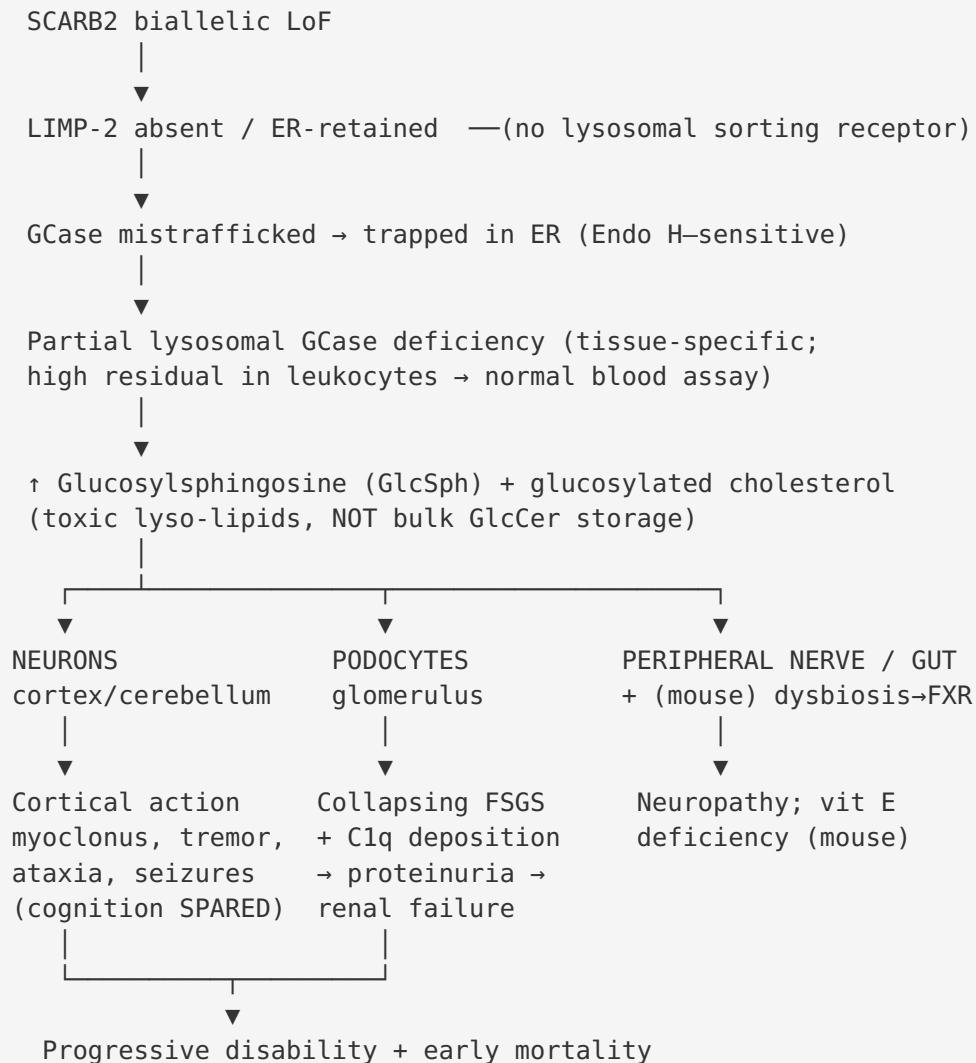
Because AMRF is Mendelian with no modifiable environmental cause, there is no primary prevention. Risk reduction relies on genetic counseling of at-risk and consanguineous families, carrier testing, cascade testing of relatives, and prenatal or preimplantation genetic diagnosis once the familial SCARB2 variants are known (carriers being unaffected per PMID: 18424452). **Secondary prevention** consists of early urinalysis surveillance for proteinuria to enable timely renal management. **Tertiary prevention** includes dialysis/transplant, aspiration precautions, antimyoclonic therapy, and—based on mouse data—vitamin E supplementation for the malabsorption-driven deficiency: "*supplementing vitamin E ameliorates the neuromotor impairment and neuropathy in Scarb2 knockout mice*" (PMID: 38635907).

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic loss-of-function variant in SCARB2** (nonsense, frameshift, splice-site, or missense) *leads to* absence, truncation, or misfolding of the LIMP-2 protein. (*Demonstrated — F001, F012*)
2. Mutant/absent LIMP-2 *results in* retention of LIMP-2 in the **endoplasmic reticulum** and failure of the LIMP-2–GCase complex to traffic to the lysosome. (*Demonstrated — F012*)

3. ER-retained, unescorted **β -glucocerebrosidase (GCase)** is **mistargeted** and remains Endo H-sensitive in the ER instead of maturing in the lysosome. (*Demonstrated in patient fibroblasts — F004, F012*)
4. Mistrafficking *results in* **partial, tissue-specific lysosomal GCase deficiency** — severe in brain, kidney, and fibroblasts, but with high residual activity in leukocytes. (*Demonstrated — F004, F010*)
5. Reduced lysosomal GCase *leads to* dysregulated glycosphingolipid catabolism, with accumulation of the toxic lyso-lipid **glucosylsphingosine (GlcSph)** and **glucosylated cholesterol**, rather than bulk glucosylceramide storage. (*Demonstrated in mouse tissue — F004; toxicity to human tissue inferred*)
6. Glycosphingolipid dysregulation and lysosomal dysfunction *result in* cellular injury that **branches** by tissue:
 7. **6a (neurological arm):** Injury to cortical/cerebellar neurons *leads to* cortical hyperexcitability → **cortical action myoclonus, tremor, ataxia, photosensitive seizures**, with intracellular inclusions and extraneuronal pigment. Cognition is spared. (*Demonstrated clinically/pathologically — F005, F013*)
 8. **6b (renal arm):** Injury to **podocytes** *leads to* **collapsing focal segmental glomerulosclerosis** with C1q deposition → proteinuria/nephrotic syndrome → **end-stage renal failure**. (*Demonstrated — F002, F013*)
 9. **6c (peripheral/systemic arm, partly from mouse):** *Leads to* peripheral demyelinating/axonal neuropathy; and (mouse) gut dysbiosis → FXR hyperactivation → lipid malabsorption → **vitamin E deficiency** compounding neuromotor deficits. (*Demonstrated in mouse; human relevance inferred — F003*)
10. Progressive neuronal and glomerular damage *results in* **cumulative disability and early mortality** — historically from untreated renal failure, now predominantly from neurological decline once dialysis/transplant manages the kidney. (*Demonstrated — F011*)



Upstream vs downstream

- **Upstream (initiating):** SCARB2 mutation → LIMP-2 loss → ER retention → GCase mistrafficking. These are the shared, obligate steps.
- **Midstream (biochemical):** partial GCase deficiency → GlcSph/GlcChol accumulation.
- **Downstream (phenotypic, branched):** neuronal injury (myoclonus/ataxia/seizures), podocyte injury (FSGS/renal failure), peripheral nerve and systemic/nutritional effects. The **renal arm is incompletely penetrant** (F006), explaining pure-PME cases; **C-terminal variant location modulates age of onset** (F006), the only established genotype–phenotype modifier.

Cell types and biological processes

Level	Entity	Ontology suggestion
Biological process	Lysosomal protein transport / GSL catabolism	GO:0007041, GO:0006687
Biological process	ER retention / protein misfolding	GO:0034976
Cell type	Cortical/cerebellar neuron	CL:0000540
Cell type	Podocyte	CL:0000653
Cell type	Schwann cell / peripheral nerve	CL:0002573
Compartment	Lysosome	GO:0005764
Compartment	Endoplasmic reticulum	GO:0005783
Metabolite	Glucosylsphingosine	CHEBI:88431

Evidence Base

PMID	Study (short title)	Type	Supports finding(s)	Contribution
18308289	<i>Array-based gene discovery... SCARB2/LIMP-2</i>	Human genetics	F001, F002, F013	Maps AMRF to 4q13-21; identifies SCARB2 as causal; defines renal pathology
18424452	<i>Nonsense mutation in LIMP-2 gene...</i>	Human genetics/biochem	F001, F002, F009, F010, F012	W178X mutation; recessive segregation; C1q nephropathy; fibroblast vs leukocyte assay; ER localization of GCase
12620969	<i>LIMP-2/LGP85 deficiency...in mice</i>	Mouse model	F003, F013	Hydronephrosis, deafness, peripheral demyelinating neuropathy
38635907	<i>Gut dysbiosis... Scarb2 deficiency</i>	Mouse model	F003, F007, F015	FXR/vitamin E axis; therapeutic rescue
40639771	<i>LIMP-2 deficiency... glycolipid abnormalities</i>	Mouse biochem	F004	GlcSph/GlcChol accumulation; high leukocyte residual GCase
15364701	<i>AMRF: characterization of a unique cerebro-renal disorder</i>	Human clinical series (n=15)	F005, F007, F011, F013, F014	Ages of onset; renal timeline; prognosis; anatomy; rarity
22050460	<i>PME without renal failure caused by SCARB2</i>	Human clinical/neurophysiology	F005, F006, F010, F011	Preserved cognition; cortical myoclonus; renal-sparing spectrum
35346091	<i>Genotype-Phenotype correlations of SCARB2</i>	Review	F006, F014	C-terminal variant location affects onset age
34337151	<i>Miglustat Therapy for AMRF</i>	Case report	F007, F011	SRT halts myoclonus; early mortality framing

PMID	Study (short title)	Type	Supports finding(s)	Contribution
21782476	<i>PME with nephropathy C1q due to SCARB2</i>	Case report (2 sibs)	F002, F007	SRT improvement; C1q nephropathy
17485520	<i>LIMP-2...cardiac intercalated disc</i>	Mouse model	F008	Non-lysosomal cardiac role
24389070	<i>LIMP-2: new player in lysosome pathology</i>	Review	F008	Enterovirus receptor; pleiotropy
19933215	<i>Disease-causing LIMP-2 mutations... binding to β-GC</i>	In vitro	F012	ER retention; coiled-coil binding domain
22537104	<i>Critical histidine... pH-sensitive binding</i>	In vitro	F012	pH-dependent V-ATPase release mechanism
33343627	<i>Distinctive EEG Patterns in SCARB2 PME</i>	Case report	F010	Fixation-off phenomenon; REM parasagittal spikes
33772352	<i>Novel homozygous splice-site in SCARB2</i>	Case report	F010	Molecular confirmation via WES (c.423+1G>A)
37529812	<i>AMRF Case Report with Bioinformatic Annotations</i>	Case report	F013, F014	Consanguinity; polyneuropathy

Evidence source distinction: Human clinical (18308289, 18424452, 15364701, 22050460, 35346091, 34337151, 21782476, 33343627, 33772352, 37529812); mouse model (12620969, 38635907, 40639771, 17485520); in vitro (19933215, 22537104); computational/constraint (gnomAD for F009).

Section-by-Section Data Summary

1. Disease Information

AMRF is an autosomal recessive cerebro-renal lysosomal disorder combining progressive myoclonus epilepsy with collapsing FSGS/renal failure. Identifiers: MONDO:0009699; OMIM #254900 (EPM4); ORPHA:163703. Synonyms: EPM4; progressive myoclonic epilepsy 4 with or without renal failure; myoclonus-nephropathy syndrome; action myoclonus-renal failure syndrome. Information is aggregated from disease-level resources plus small clinical case series/reports (individual patients).

2. Etiology

Primary cause is purely genetic: biallelic LoF variants in *SCARB2* (F001, F014). Genetic risk factor: two pathogenic *SCARB2* alleles; consanguinity elevates risk (F014). No environmental, infectious, toxic, lifestyle risk or protective factors are established. C-terminal variant location is the only identified modifier of onset (F006). No meaningful gene-environment interaction is documented in humans, though the mouse gut-dysbiosis/vitamin-E axis (F003) hints at a diet-modifiable secondary pathway.

3. Phenotypes

Phenotype	Type	HPO	Onset	Progression	Frequency
Action myoclonus	Clinical sign	HP:0001336	mean 21.7 y	Progressive/ disabling	Near-universal
Tremor	Clinical sign	HP:0002345	mean 19.8 y	Progressive	Common (often first)
Ataxia / cerebellar signs	Clinical sign	HP:0001251	Young adult	Progressive	Common
Generalized seizures	Clinical sign	HP:0002197	mean 22.7 y	Episodic	Most cases (infrequent seizures)
Photosensitivity	Neurophysiologic	HP:0007000	Young adult	—	Pronounced
Proteinuria/ FSGS	Lab/pathology	HP:0000097	ages 9–30	Progressive → renal failure	~universal in AMRF; absent in renal-sparing PME
Peripheral neuropathy	Clinical sign	HP:0009830	Variable	Progressive	Variable
Preserved cognition	(Negative)	—	—	Stable	Characteristic

QoL impact is severe due to disabling myoclonus, dysphagia, and dialysis dependence.

4. Genetic/Molecular

Causal gene *SCARB2* (HGNC:10534; OMIM *602257). Variant types: nonsense (W178X), frameshift (p.N45MfsX88), splice-site (c.423+1G>A), missense—all loss-of-function; classified pathogenic/likely pathogenic under ACMG. gnomAD: LoF-tolerant (LOEUF 0.67), consistent with recessive inheritance (F009). Germline origin. Functional consequence: loss of function via ER retention (F012). No established modifier genes beyond variant position; no epigenetic mechanism or chromosomal abnormality implicated.

5. Environmental

Not applicable—no environmental, lifestyle, or infectious cause. (Note: LIMP-2 is an enterovirus receptor, but this is unrelated to AMRF causation — F008.)

6. Mechanism

See Mechanistic Model above (F004, F012, F013).

7. Anatomical Structures

Primary organs: brain (cerebral/cerebellar cortex — UBERON:0000956/0002129) and kidney (glomerulus — UBERON:0000074). Secondary: peripheral nerve (UBERON:0000044). Cells: neurons (CL:0000540), podocytes (CL:0000653). Subcellular: lysosome (GO:0005764), ER (GO:0005783). Bilateral/symmetric involvement (F013).

8. Temporal Development

Onset: adolescence–young adulthood, insidious/chronic. Course: relentlessly progressive; proteinuria → renal failure within 0–8 years; myoclonus progressively disabling. No spontaneous remission. Lifelong (F005, F011).

9. Inheritance and Population

Autosomal recessive; ultra-rare (<1/1,000,000); ~1:1 sex ratio; consanguinity/founder effects in some kindreds. Carriers unaffected (F009, F014). Penetrance of the neurological phenotype is high; renal penetrance is incomplete/variable (F006).

10. Diagnostics

Cortical-myoclonus electrophysiology (giant SEPs, photosensitivity, EEG–EMG coherence, fixation-off phenomenon, REM parasagittal spikes); urinalysis/renal biopsy (collapsing FSGS, C1q); molecular *SCARB2* testing (WES/panel/single-gene) as gold standard; fibroblast (not leukocyte) GCase assay (F010).

11. Outcome/Prognosis

Progressive, disabling, early mortality; historically fatal from renal failure, now neurological. Cognition preserved. No 5-year survival figures established for this ultra-rare disease (F011).

12. Treatment

Symptomatic: antiepileptic/antiseizure drugs (valproate, levetiracetam, piracetam, clonazepam, perampanel), renal replacement therapy. Emerging: SRT with miglustat (F007), vitamin E repletion (F007, F015). No approved disease-modifying therapy.

13. Prevention

Genetic counseling, carrier/cascade testing, prenatal/PGD; proteinuria surveillance; dialysis/transplant; vitamin E (F015).

14. Other Species / Natural Disease

Mouse (*Mus musculus*, NCBI:txid10090) *Scarb2* ortholog; knockout models are the principal natural/experimental analog (F003). No naturally occurring companion-animal AMRF documented in the reviewed literature.

15. Model Organisms

Mouse *Scarb2/Limp2* knockout — recapitulates peripheral neuropathy, hydronephrosis, deafness, cortical inclusions, and (with aging) vitamin-E-deficient neurodegeneration; validated therapeutic rescue with vitamin E/FXR inhibition (F003). Limitation: mice show prominent hydronephrosis/deafness not central to human AMRF and less pronounced glomerulosclerosis, so the model captures the neurological and lysosomal biology better than the human collapsing-FSGS renal phenotype.

Limitations and Knowledge Gaps

- 1. No natural history cohort with survival statistics.** Because AMRF is ultra-rare (only dozens of reported cases), there are no Kaplan–Meier survival curves, formal prevalence/incidence figures, or validated prognostic models; prognosis is inferred from small series and case reports.
- 2. Renal phenotype in the mouse is incomplete.** The knockout emphasizes hydronephrosis, deafness, and neuropathy rather than the collapsing FSGS central to human AMRF, limiting mechanistic study of the podocyte injury pathway.

3. **GlcSph toxicity is demonstrated in mouse tissue, not directly in human AMRF brain/kidney.** The step linking GlcSph accumulation to neuronal/podocyte death is inferred for humans (F004).
4. **Therapeutic evidence is anecdotal.** Miglustat and vitamin E data derive from single case reports and mouse studies; no controlled trials exist.
5. **Incomplete penetrance of the renal arm is unexplained** beyond the C-terminal variant correlation (F006); the modifiers determining whether a patient develops renal failure are unknown.
6. **Epigenetic, transcriptomic, proteomic, and single-cell profiling of human AMRF tissue is essentially absent** from the reviewed literature.

Proposed Follow-up Experiments / Actions

1. **Establish an international AMRF registry** to derive natural-history data (age-specific survival, renal-vs-neurological mortality, penetrance of the renal phenotype).
 2. **Podocyte-specific or conditional *Scarb2* knockout / patient iPSC-derived podocyte and cortical-neuron models** to dissect the branched neuro-renal injury and test whether GlcSph reduction rescues each cell type.
 3. **Prospective trial (or n-of-1 protocols) of miglustat SRT** in genetically confirmed AMRF, with GlcSph as a pharmacodynamic biomarker in CSF/plasma.
 4. **Test vitamin E repletion and FXR modulation in human patients**, translating the mouse gut-dysbiosis/malabsorption finding, with serum vitamin E and neuropathy endpoints.
 5. **Systematic genotype–phenotype study** correlating variant position (especially C-terminal), residual GCase activity, and GlcSph levels with renal-arm penetrance and onset age.
 6. **Develop plasma GlcSph as a diagnostic/monitoring biomarker**, potentially resolving the leukocyte-assay pitfall and enabling earlier diagnosis.
 7. **Cascade carrier screening and reproductive counseling protocols** for consanguineous kindreds with known *SCARB2* variants.
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Report compiled from 15 confirmed findings across 5 investigation iterations and 36 reviewed papers. Evidence types are distinguished as human clinical, mouse model, in vitro, and computational throughout.

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